

craft, it is assumed to be a hostile aircraft, and its course is continuously tracked. All air-surveillance radars of the CSG have IFF.

3D air-surveillance radar. If a target of interest appears, 3D radar AN/SPS-48 takes over. A 3D radar provides the range and bearing like the 2D radar but also provides the height of the target, which is an extra dimension of the target's position.

Normally, to find the height of the target, a separate radar known as a height-finding radar (HFR) is used with the 2D radar. HFR's antenna always nods up and down to direct the beams to find the height of the target.

But 3D radar is a computer-controlled planar-array radar that varies its frequency instead of nodding the antenna up and down. By changing the frequency of the transmitted RF energy, this radar radiates a series of pencil beams. If the transmitted frequency is increased, the beam travels down the face of the antenna, and vice versa. The radar computer keeps track of the frequencies as these are transmitted and detects when the target reflects these. Further, it converts returned frequencies into meaningful altitude data. For finding the range



Fig. 4: AN/SPS-48 radar antenna

and bearing, it simply rotates the antenna like a 2D radar.

Operating in the range of 2900MHz to 3100.5MHz (S band), a 3D radar can detect targets up to a range of 407km (220 nautical-miles) and heights up to 30km. It emits radar energy with a peak power of 2.2MW and an average power of 35kW. It has automatic detection and tracking capabilities to simultaneously detect and track multiple air targets. It also has a moving target

indicating (MTI) capability, which helps distinguish between moving targets from stationary targets. Hills or a landmass produce radar echoes, which is landmass clutter. In the radar-signal-processing circuitry, echoes from stationary features are attenuated. The resultant echo is of the moving targets.

Radar AN/SPS-48 is electronically-stabilised instead of gyro-stabilised. To compensate for the pitch and roll of the ship, the frequency emitted is changed to compensate it. This is similar to the height-finding technique.

Using this tracking data, fighter aircraft can be scrambled towards incoming threat to intercept it. The carrier constantly updates the aircraft with the target's location until interception, through datalinks.

However, if the enemy aircraft has launched its cruise missiles, the cruise missiles become the target. Engaging the cruise missile through the fighter aircraft becomes difficult. So, air defence warfare progresses to Phase III and is carried out through surface-to-air missiles (SAMs) of the CSG.

Phase III and phase IV of air defence operations will be covered in next part of this article.

To be continued next month

What is IFF and why is it necessary

During wartimes, under combat stress, identification of friends among enemies becomes difficult. A significant portion of fatalities occur due to mistakenly identifying a friendly unit as enemy. This is called friendly fire.

A notable incident occurred during the 1971 India-Pakistan war. After inflicting terrible damage to Karachi Harbour, Indian missile boats withdrew deep into Arabian Sea. Combat aircraft of Pakistan Air Force searching for these saw a warship and engaged it. But to the horror of both parties, that ship was a Pakistani warship searching for escaped Indian missile boats.

Identifying friends is important but, at the same time, enemies also have to be identified. During World War II, in Battle of Coral Sea, American combat aircraft intercepted some Japanese carrier based aircraft and shot down many of these. Remaining Japanese aircraft withdrew and headed for their carriers. These used radio direction finders (RDFs) to pin-point the location of their carriers.

In an RDF, there is a transmitter in the carrier continuously transmits a signal. The aircraft has a radio, which detects this signal and gives the exact direction of the transmitter.

Japanese aircraft were flying back in the evening towards their carriers. Unfortunately, unknown to them, Japanese transmitter's frequency was very close to that of the Americans. When the Japanese saw a carrier, they flew right beside it. The carrier flashed signal lights to the aircraft in an effort to confirm their identity. To the horror of the Japanese, it was an American carrier and not Japanese. The men in the carrier saw that those aircraft were not responding. This turned into a nightmare for the Americans as they realised that those were Japanese aircraft and not theirs.